

Implementing a Digital Strategy: LEARNING FROM THE EXPERIENCE OF THREE DIGITAL TRANSFORMATION PROJECTS

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SUMMARY

The rapid growth of digital technologies and the extraordinary amount of data that devices and applications collect each day are increasingly driving companies to radically transform the business architecture through which they create and appropriate value. However, companies may fail to extract value from digital transformation due to the disconnection between strategy formulation and strategy implementation. Through the analysis of three case studies of firms that digitally transformed their business—namely ABB, CNH Industrial, and Vodafone—this article presents a framework that can help companies implement their digital transformation strategy and thereby renovate their business model.

KEYWORDS: digital transformation, digital strategy, strategy implementation

“Many companies define great digital transformation strategies, but there is a huge difference between having a well-reasoned digital strategy on paper and successfully implementing it . . . Most digital transformation projects fail due to poor strategy execution.”

—Adriano Gerardelli, Director of Digital Strategy & Innovation,
PricewaterhouseCoopers (PwC)

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The remarkable growth of digital technologies and the increasing pervasiveness and reliability of high-speed Internet services have radically reshaped the operations and business models of companies.¹ This has led to substantial changes in their activities, processes, and capabilities.² A growing number of companies have adopted a digital transformation strategy to transform how they create and appropriate value. Consequently, they need to reexamine and revise the current architecture of their value creation and appropriation models to sustain their competitive advantage. Also, a key requirement for firms adopting digital transformation is to renovate their business models so that they are consistent with their business strategy.³

New digital technologies can improve competitive performance by increasing the flexibility of products and services by supporting the continuous evolution of their scope, features, and value, even after they have reached the market;⁴ lowering the barriers across industries, favoring connections, exchanges, and partnerships among companies operating in different sectors;⁵ and supporting companies in accessing continuous, timely, and reliable data streams.⁶

Digital transformation can lead to notable advantages for firms, such as helping create products and services that are more efficient and consistent with customer needs,⁷ providing a shorter innovation process and time to market,⁸ and creating related digital ecosystems.⁹ Moreover, digital transformation favors the interconnection among diverse industries by guiding firms to new opportunities for creating and appropriating value through digitization and connectivity.¹⁰ For instance, Becton Dickinson, a medical equipment manufacturer, has been developing connections with software and analytics industries to increase the effectiveness of its products.¹¹ Also, another example of how digital transformation reduces barriers between industries is that companies such as Google, Apple, and Uber are devoting more attention to the automotive industry for the development of autonomous vehicles.

However, the adoption of digital transformation strategies also involves challenges.¹² According to recent estimates, 66% to 84% of digital transformation projects fail,¹³ which is a sizable proportion considering the costs, both monetary and otherwise, of putting these projects in place. One major challenge is to ensure consistency between strategy formulation and strategy implementation,¹⁴ which despite their interdependence are considered distinct concepts. Specifically, digital strategy formulation refers to defining a guiding policy for the creation and appropriation of value by exploiting digital technologies to achieve long-term objectives—which include factors related to the external environment, the technological potential in the current competitive scenario, and the evolution of the market. Therefore, digital strategy formulation should identify the elements of the firm's business model that must be modified according to the new strategy, along with the scope of the digital transformation.

In contrast, digital strategy implementation refers to how firms translate the digital strategy formulated into a concrete plan and set of actions.¹⁵ Careful implementation is crucial to ensure consistency between the firm's actions and

the objectives defined in the digital strategy formulation.¹⁶ In fact, existing knowledge implicitly assumes that once a strategy has been defined, implementation will follow.¹⁷ However, corporate practice shows that this is *not* always the case, and senior executives cannot achieve any benefit from digital transformation strategies if they cannot effectively implement them.¹⁸ Prior studies point out that effective strategy implementation is more critical for avoiding failure as compared with good strategy formulation.¹⁹ On the one hand, a precise implementation allows for adaptation to evolving conditions,²⁰ thus correcting for an inaccurate formulation,²¹ but on the other hand, a good formulation is of no value if not properly executed.²²

A prominent example of this is GE. GE's top management planned to digitally transform the firm. However, while deemed an appropriate strategic choice in the current competitive environment, GE failed to implement this strategy to the point of having to fire more than 100 employees at the software operation facility that had been created to support GE's digital strategy.²³ Recently, John Flannery, GE's CEO, pointed out that the company is "still deeply committed to [digital], but we want a much more focused strategy."²⁴ Given such disconnection between strategy formulation and implementation,²⁵ our aim is to understand *how* firms can implement a digital transformation strategy. Specifically, we describe and analyze three cases of firms that digitally transformed their business—ABB, CNH Industrial, and Vodafone. They were supported by a "digital companion" globally renowned for its excellence in the execution of digital transformation strategies, namely, Microsoft. The rich body of qualitative evidence allowed us to identify the critical building blocks—resources, capabilities, and activities, as well as stakeholders—that need to be taken into account when a firm implements a digital transformation strategy. Our framework illustrates how the three companies used these building blocks to ensure consistency between their strategy formulation and implementation, which led to successful digital transformation projects. In particular, our framework allows companies to renovate their business models, conceived as the "conceptual and architectural implementation of a business strategy."²⁶

Conceptual Background

The increasing spread of new digital technologies is disrupting existing industries.²⁷ Indeed, due to digitalization, many products and services offer new features and functions. A prominent example is the Nest thermostat, which alongside the traditional functions increases energy use efficiency by collecting data on energy consumption, sharing these with utilities for more accurate forecasting, improving the service, providing customers with suggestions to reduce energy consumption, and connecting to other home devices.²⁸ Another notable example of a company that has completely and successfully revised its business thanks to digitalization is Netflix. Originally, Netflix was an online digital video disk (DVD)-by-mail sales and rental store. However, consequent to the boost in data connection speed and its lower costs, as well as improvements in

video-on-demand service effectiveness and efficiency, Netflix digitally transformed its competitive strategy by offering a worldwide video streaming service, exploiting data on movie consumption to understand major trends in the entertainment business, and eventually becoming an original content producer.²⁹ As these two examples show, the pervasive use of digital technologies and the ability to collect consumption and utilization data enable companies to rewire their traditional business model into a digital business model³⁰ that can lead to increasing their competitive advantage.

However, digital transformation is not always straightforward. Indeed, due to the disruption in activities, processes, and capabilities, digital transformation processes often fail.³¹ Despite the appropriateness of adopting a digital transformation strategy, the outcomes may be far from those expected. For instance, as in the GE case, Nike failed to reap the benefits of digital transformation with its Nike+ personal fitness products.³² Nike+ products incorporated sensors that collected data on customer activities, and synchronized these through a web platform. In this way, customers could receive feedback and suggestions to improve their physical performance, along with the possibility to access a virtual community of friends, athletes, and coaches.³³ In turn, Nike could collect data on customers (including their activities and preferences) to fine-tune its marketing activities.³⁴ However, while the digital transformation project was promising, Nike discontinued its Nike+ products³⁵ and only recently attempted to apply digital technologies to achieve a different objective.³⁶

There are many reasons why digital transformation projects fail, including the failure to consider important aspects of change management in relation to employees and customers who are required to change their way of working and interacting with the brand. Developing a proper strategy for effectively leveraging digital technologies is crucial for the success of digital transformation projects.³⁷ Defining a digital transformation strategy corresponds to developing a plan of action to achieve a specific goal³⁸ through the strategic renewal of the firm.³⁹ However, despite formulating an appropriate digital transformation strategy, companies often fail in implementing the strategy because implementation is particularly risky and uncertain when companies have to deal with a disruptive change to their business following the introduction of new digital technologies.⁴⁰

In order to effectively support the actual implementation, firms need to rely on business models that reflect each, individual firm's strategy.⁴¹ In fact, business models are a conceptual tool used to depict how firms create and appropriate value, adapting the previously defined strategy to the contingencies that actually take place.⁴² Hence, they represent a logical structure for the linkage between the formulated strategy and its contingent implementation.⁴³ Business models describe the elements and the relationships leveraged by firms to create and appropriate value.⁴⁴ They are made of four main components: the firm's value proposition and market segments; the structure of the value chain; the mechanisms used by the firm to appropriate the value provided; and the relationships

among these elements.⁴⁵ They provide a complete description of a firm's strategy⁴⁶ and can be helpful when companies need to thoroughly revise their strategies due to disruptive changes.⁴⁷

A digital transformation may require a substantial change in the business model to reap the maximum advantages and to reduce the cost burden.⁴⁸ In a digital transformation strategy, the role of the business model is essential in adapting the architecture of a firm's value proposition, market segments, value chain, and value appropriation to emerging contingencies.⁴⁹ The introduction of digital technologies calls into question the traditional way of doing business,⁵⁰ and companies must thus reconsider which elements to leverage to establish and sustain their competitive advantage.⁵¹ Identifying the core aspects behind a digital strategy and its implementation allows companies to apply the digital lens to their current business, ascertain new modes of value creation, evaluate new ways of value appropriation,⁵² and consequently renew their strategies.⁵³ For instance, while data streams are paramount for firms adopting digital technologies,⁵⁴ traditional business model frameworks⁵⁵ do not assign them the central role they have in supporting the digital transformation.

Research Design and Methodology

For our research, we adopted a case study methodology⁵⁶ along with the principles of engaged scholarship.⁵⁷ One of the authors is a Microsoft manager who directly followed several digital transformation projects on behalf of her company. Notably, Microsoft has, in recent years, increasingly partnered with companies wishing to transform their businesses by leveraging data and technologies, thus becoming an influential player in the digital transformation ecosystem. Acknowledging the difficulties that enterprises may face in embarking on digital transformation processes, Microsoft has positioned itself as a partner aiming to accompany firms along this journey by becoming a "digital companion" for organizations embracing a digital transformation strategy. Satya Nadella, Microsoft's CEO, explained this strategic vision, noting,

Companies are focused on ensuring that they stay relevant and competitive by embracing this [digital] transformation. And we want Microsoft to be their partner. To do so, there are four initiatives every company must make a priority. The first is engaging their customer base by leveraging data to improve the customer experience. Second, they must empower their own employees by enabling greater and more mobile productivity and collaboration in the new digital world of work. Third, they must optimize operations, automating and simplifying business processes across sales, operations, and finance. Fourth, they must transform their products, services, and business models.⁵⁸

Given the vision and commitment to partnering with firms to help them digitally transform their businesses, the three companies that Microsoft accompanied provide an opportunity to observe how digital transformation strategies are implemented with the assistance of an expert companion. Indeed, Microsoft's

experience was fundamental to identifying the specific building blocks that constitute the framework we constructed from our analysis.

For our sampling strategy, we selected digital transformation projects that had involved Microsoft and can be considered exemplars for the implementation of a digital transformation strategy.⁵⁹ The author who is a Microsoft manager critically revised the portfolio of projects she has participated in during the last five years with the idea to build a polar-type sample, which included both successful and unsuccessful cases of digital strategy implementation. The idea was to compare implementation projects that delivered positive results with those that were unsuccessful to spot differences and unearth factors linked with successful digital strategy implementation more easily. Unfortunately, due to privacy and confidentiality reasons, it was not possible to have access to the data required to carefully study the unsuccessful cases. Therefore, we decided to focus on cases that were illustrative examples of a successful alignment between strategy formulation and implementation in different contexts (in particular, manufacturing and service companies) to allow for potential differences among cases. Following these criteria, the case selection brought to our attention three organizations that successfully implemented a digital transformation strategy (ABB, CNH Industrial, and Vodafone), allowing us to highlight the critical elements to take into account to effectively implement a digital transformation strategy. Thanks to the direct involvement in the realization of these projects by one of the authors, we have had privileged access to data and information that were especially useful to inductively build a model of digital strategy implementation.

Data from the cases were collected using primary and secondary sources. In particular, our Microsoft-affiliated author was directly involved in the digital transformation implementation in the three cases, working on the execution of these projects for an average period of 12 months each, consistent with the engaged scholarship methodology.⁶⁰ She had the opportunity to take part in the projects and access primary sources of information, such as aggregated data, internal archival records and reports, and interviews with those involved in the digital transformation processes. In particular, the interviews were based on a structured list of questions designed to provide a clear understanding of the digital transformation processes and the business model renewal. Moreover, interviewees were encouraged to share further insights that could support the research team to get a clearer picture of the processes. Furthermore, secondary sources, including corporate websites and business magazine articles, provided a clear picture of the processes. The data were gathered in 2018 and refer to the period 2016-2018. Table 1 reports some general information on the cases.

The researchers then analyzed the data collected following an inductive approach. The authors independently reviewed the cases to identify the building blocks supporting the implementation of a digital transformation strategy. In the first phase, each author coded and labeled the transcripts of the interviews and other primary and secondary sources documents in order to highlight features

TABLE I. Overview of the Three Case Studies.

	ABB	CNH Industrial	Vodafone
<i>Business</i>	Electrical equipment	Agricultural equipment	Telecommunications
<i>Size</i>	147,000 employees (2018)	63,000 employees (2017)	111,000 employees (2018)
<i>Headquarters</i>	Zurich (Switzerland)	Amsterdam (The Netherlands)	London (United Kingdom)
<i>Founded</i>	1988	2012	1991
<i>Main objective of the digital transformation project</i>	Develop smart products that allow providing value-added services to customers	Develop autonomous unmanned agricultural machines endowed with AI	Improve customer care services using conversational autonomous interfaces based on AI operating through a number of channels (web, apps, social networks, etc.)
<i>Start of digital transformation project</i>	2016—year of ABB Ability™, under which ABB consolidated its digital solutions	2017	2017
<i>Number of employees involved</i>	Undisclosed	30 employees including Commercial Vehicles and Industry-specific Vehicles Unit Managers and IT, Operations, and Executive Business Stakeholders	23 employees working in Commercial Operations Unit and five employees working in IT

Note: AI = artificial intelligence; IT = information technology.

related to the business model transformation of each case and referring to typical business model elements.⁶¹ Afterward, the results of this phase were compared across cases to spot similar patterns and emerging concepts.⁶² Then, the individual outputs were discussed and synthesized during ad hoc meetings to construct the proposed framework. In particular, the definition of the business model elements to be included in the proposed framework was consistent and representative with regard to the theoretical understanding about business model elements.⁶³ During this phase, the emerging results were continuously compared with the literature to spot confirming and conflicting findings with respect to the extant knowledge. In this way, we corroborated the internal validity and increased the generalizability of our results.⁶⁴ Finally, we asked our key informants whether they felt the framework was a reasonable description of what had occurred at the companies in light of their own experiences, and all agreed.

Overview of the Three Cases

ABB

Established in 1988 after the merger of Sweden's ASEA and Swiss Brown Boveri and Cie, ABB is a Swiss/Swedish company that operates in power and automation technology development with utilities and industrial firms as customers. Its roots go back more than 130 years. ABB has been driving digital transformation since the early 1970s when ASEA introduced the world's first microprocessor-controlled robot. This process resulted in the launch of the ABB Ability™ brand in 2016, under which ABB consolidated its digital transformation solutions, focused on creating additional value for customers by providing software-enabled services. One of ABB's objectives was to offer a pay-per-use service for specific devices, and to do so, it created digital solutions to continuously sense the state of devices and offer digital support services, such as predictive maintenance, forecasting, and optimization.

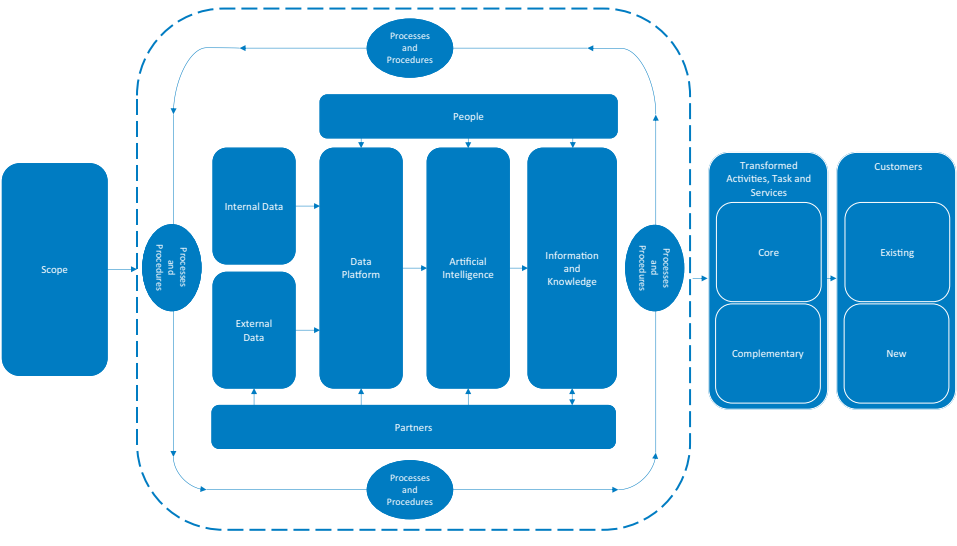
CNH Industrial

CNH Industrial, registered in the Netherlands with corporate offices in London, was founded in 2012 following the demerger of Fiat's nonautomotive businesses. Previously, these had been run as two separate Fiat-owned business units, Fiat Industrial and CNH Global. CNH Industrial's core business is the design and production of agricultural and construction equipment, commercial vehicles, and powertrains. CNH Industrial also offers financial services to its customers. The digital transformation project under study focused on the agricultural equipment business. In particular, CNH Industrial is committed to driving the evolution of the agricultural industry, supporting the development of the digital farming paradigm. Specifically, CNH Industrial aims to connect all the stages of farming through a digital platform to offer automation capabilities, value-added services, connecting customers with internal and external partners, and promoting a servitized business model. The digital transformation project we analyzed began in 2017 and involved developing autonomous unmanned agricultural machines, endowed with artificial intelligence (AI), that operate through a digital platform. In total, 30 employees were involved in the project, including managers from the commercial vehicles unit, industry-specific vehicles unit, as well as information technology (IT), operations, and executive business stakeholders.

Vodafone

Vodafone is a United Kingdom-based company founded in 1991 and operating in the telecommunications industry. Vodafone is a mobile operator present in 25 countries in Europe, Africa, Asia, and Oceania, while also covering the Americas with partnerships. Vodafone's digital transformation project began in 2017 and was focused on improving its customer care services, deemed critical to retaining customers. Specifically, Vodafone's objectives were cost reduction, customer care process optimization, and improving digital interaction with customers through AI. Accordingly, Vodafone leveraged Microsoft's digital services to develop conversational autonomous interfaces based on neural networks

FIGURE 1. The digital strategy implementation framework.



processing natural language, and able to interact with customers through several channels (e.g., voice, apps, social networks, websites, and home assistants). The project initially involved five employees from the IT unit and 23 from the commercial operations unit. Moreover, thanks to a successful process of change management and new operating model implementation, the project has now added several digital hubs that operate following agile methodologies in developing their new digital products with an incremental and iterative approach. This digital transformation project is redefining the way Vodafone listens, understands, and assists end customers, which, after rolling out in Italy, will be extended to other countries in which Vodafone operates.

Findings

The analysis of the three case studies allowed us to construct a framework that companies can use to effectively implement their digital transformation strategies (Figure 1).

Our cases suggest that the starting point of the effective implementation of a digital transformation strategy is defining the scope of the transformation.⁶⁵ Clearly defining what a company wants to achieve is critical to maintaining focus on the digital transformation goal and ensuring the consistency of each building block with the strategy formulated. In fact, a major output of the strategy formulation process is the definition of the organization's strategic goals.⁶⁶ A crucial element of a digital transformation strategy, and consequently of our framework, is data. Indeed, data have a central role in the digital economy⁶⁷ and are an enabler of digital transformation.⁶⁸ The most important aspect of data usage is that it must be constantly refreshed. New data need to be continuously collected to support the analyses and

data models in a feedback loop. Once collected, cleaned, and securely stored, data are then ready to be processed through specific AI techniques⁶⁹ to extract information that feeds the organizational knowledge base. Companies must define the relevant job roles,⁷⁰ the strategic partners, and the processes and procedures⁷¹ needed to support the information extraction and knowledge generation process. Then, the information and knowledge generated is used to carry out and support the transformed activities, tasks, and services that create value for customers.⁷²

Scope

To be effective and avoid inefficiencies, companies must have the scope of the digital transformation strategy clearly in mind.⁷³ This is the cornerstone of defining how the company envisions creating value for its customers. Such scope is defined on the basis of the strategic goals resulting from the strategy formulation process and favor the connection between strategy formulation and implementation. In the three cases, the scope of the digital transformation was clearly defined as follows:

- ABB: Create continuous value for customers through software- and platform-enabled services.
- CNH Industrial: Develop new services around predictive maintenance and intelligent logistics through the digitalization of its fleet.
- Vodafone: Automate and improve customer care.

CNH Industrial and ABB sought to change their business by creating digital platforms that collect data and leverage data to enable new, high added-value services for their customers. Vodafone sought to enhance the value of extant services, such as customer care, by leveraging digital technologies.

Data Sources (External and Internal)

Properly managing data is critical to effectively support the digital transformation of firms,⁷⁴ while the peculiar role of data in value creation has been sometimes overlooked and recognized only recently.⁷⁵ The three companies rely on both internal and external data sources to implement digital transformation strategies. For instance, in the ABB case, the company needed to understand how customers use its products to gain useful insights for the entire organization. To achieve this, ABB relied on internal data sources, such as the data provided by Internet of Things (IoT) devices connected with products, and external data sources, such as consultants, installers, panel builders, and original equipment manufacturers (OEMs). CNH Industrial uses sensors on products to determine the status of vehicles, while data sourced from external partners (e.g., retailers, insurance companies, and seed and fertilizer suppliers) are useful to infer additional insights. Similarly, Vodafone largely relies on internal data obtained from customer interactions, while baseline conversational models allow fine-tuning the service. Since critical resources, such as data, allow to establish and sustain the firm's competitive advantage,⁷⁶ the three companies were very careful

in ensuring their control over them. In some cases, companies need to ensure access to data sources by internalizing them (e.g., using IoT devices to guarantee a continuous data stream from products sold) or by establishing reliable agreements with external sources, such as formal partnerships.

Data Platform

Data are usually transmitted via a data platform through which the product and all the software as a service (SaaS) and platform as a service (PaaS) are generated and then pushed to end customers and other players in the greater ecosystem.⁷⁷ For example, in the ABB case, a digital platform is used to collect data from products and make them available for the knowledge extraction needed to provide high-value software-enabled services. This means that every action/input starts and ends as a digital signal that assumes various meanings based on the company's business view. In particular, data platforms operate as a place where data are collected from internal and external sources, enriched, and made available through a structured and business-oriented data library. As a result, data can be accessed by various areas of the business to create value either through data mining and AI model experimentation or through data services powering business applications and operations. For instance, CNH Industrial's data platform collects data from IoT devices and makes them available for analyses and machine-learning model creation by internal data scientists and product managers. Moreover, these platforms often collect end-user data, and must hence be accurately governed and protected in compliance with law (i.e., General Data Protection Regulation in Europe). Due to the confidentiality of the data, CNH Industrial has developed specific encryption, while Vodafone has defined internal policy and privacy guidelines to protect them.

People

Generally, digital transformation entails a thorough revision of the firm's operations and business models.⁷⁸ However, when substantially revising the activities and processes, new professional roles may be needed. On one side, firms may define a new managing role to drive the transformation (e.g., Chief Digital Officer);⁷⁹ on the other side, employees may have to possess specific skills and capabilities to fully seize the opportunities that digital technologies create.⁸⁰ CNH Industrial, alongside the Microsoft professionals working on the transformation program, supported its data scientists in developing new methodologies and programming competences. This was essential since the digital transformation project also pushed CNH Industrial to become a software developer and, ultimately, adopt an open platform model providing and selling services to third parties. Therefore, professional roles (such as digital advisors) and a new digital team were created within the CNH Industrial's existing IT organization to support the strategy implementation and execution. In the Vodafone case, the digital transformation project compelled managers of the commercial operations unit to enhance their employees' capabilities. A call-center unit was trained to no longer answer customer calls directly, but to design conversational frameworks for the chatbot to be used in serving customer requests. In addition, these employees

were involved in training the conversational models to become more and more accurate and relevant for the customers. They did this through a digital feedback loop process of continuous improvement of the accuracy and relevance of conversations, based on customers' experiences. Moreover, a neural network training unit was set up to enable operators to use the new intelligent system, which resulted in new jobs and professions. In particular, the project required employees able to train AI and conversation designers. However, in the ABB case, the approach did not impel employees to dramatically change their routines; in fact, most were able to basically do the same work as before, but with new decision support intelligence.

Partners

The digital transformation of companies may entail a radical change in their core capabilities. In the CNH Industrial case, the company has evolved from offering commercial vehicles to operating connected vehicles, therefore, requiring knowledge and competences that differed significantly from the past. Defining agreements with partners can support the organization in obtaining new data, capabilities, knowledge, and competences that are crucial for the implementation of the digital transformation strategy.⁸¹ Partnerships can be established with several types of stakeholders. ABB, CNH Industrial, and Vodafone all established a partnership with Microsoft to develop the IT infrastructure needed to sustain the digital transformation of their business. Other partnerships can also be pursued over time to support the implementation of the digital transformation strategy. For instance, CNH Industrial established partnerships with both companies and customers to obtain insights on their products and collaborate by sharing anonymized data to enhance AI models governing vehicles' digital experience. ABB created partnerships with stakeholders such as OEMs, distributors, and panel builders in order to improve its offering and support the new service design and favoring the development of products' core components. Previous studies have highlighted the role of partnerships in assisting with the revision and implementation of novel firms' digital strategies.⁸² The role of partners in our framework illustrates their connection with the other building blocks supporting the implementation of a digital transformation strategy.

Artificial Intelligence (AI)

In the three cases presented, the data collected are used to develop and test machine-learning models deployed for various purposes. Specifically, Microsoft AI technologies were adopted within a rapid insight and data exploration framework to ensure an agile approach to data discovery and value creation. To change the business model and organizational activities, lean analytics and an AI operations framework are needed. In fact, "learn fast and fail fast" is at the core of every approach to data and machine-learning model design and experimentation. This approach is a key success factor that allows for developing better solutions to existing problems, identifying new patterns in data that promote specific actions, inferring relevant knowledge, and promoting both radical and incremental improvements in products and services.⁸³ Therefore, a digital business model should define the specific AI strategies and capabilities needed to

transform data into information and, eventually, generate knowledge that can be used to create value for customers. Typical examples of AI products are applications using computer vision, facial recognition, autonomous vehicles, virtual agents, machine-learning models, natural language processing, artificial neural networks, and big data analytics. In the three companies, machine learning is currently the most relevant and diffused technique to obtain the most from data. The relevance of the AI building block is a peculiarity of the digital context that has not been fully unraveled in previous studies,⁸⁴ however, it is of core importance in our analysis of the three companies.

Information and Knowledge

This building block involves the output of the data analysis. In the case of ABB, the intelligence extracted by the data platform can help the company and various stakeholders understand how customers use the products and how the products affect the customers' business (in business-to-business [B2B] relationships). This process is similar for service companies. In the CNH Industrial case, data are taken from telematic and telemetry boxes (IoT devices) that are then sent to the cloud where they are computed, cleaned, and modeled to be subsequently forwarded to a control room that proactively uses them to globally monitor all vehicles. Thanks to predictive maintenance models, the control room can send alerts on the status of vehicles and understand how drivers interact with the monitored vehicles. This allows CNH Industrial to provide a new service to their first-party customers as well as to let them share information to their third parties in the form of data-as-a-service. In the Vodafone case, data analysis allows developing enhanced conversation models that result in reshaping the customer care operations. Finally, the extracted information may be processed to further increase individual and organizational knowledge. The importance of this building block when implementing a digital transformation strategy has also been partially neglected in previous literature on business models.⁸⁵

Processes and Procedures

The implementation of a digital transformation strategy may require companies to revise the processes and procedures they use to create value for customers, since the changes involved could be radical.⁸⁶ In fact, prior research and our empirical evidence suggest that processes and procedures should be agile and lean when dealing with digital transformation in order to allow the company to adapt to rapid change and seize emerging opportunities, thus emulating the behavior of startups rather than that of consolidated companies.⁸⁷ In addition, this building block can also involve the revision of the formal relationships among employees and the formation of dedicated business units. For instance, in the ABB case, product managers drive the business idea through the iterative development of minimum viable products to achieve quick wins. In the CNH Industrial and Vodafone cases, a similar lean approach was found. At CNH Industrial, the digital transformation project was carried out by adopting experimental and iterative approaches, lowering the barriers between developers and business owners, and thus allowing for real-time feedback cycles on

the scheduled work. These new processes and procedures supported the digital transformation strategies through timely checks and refined implementation for consistency with creating value for customers.

Transformed Activities, Tasks, and Services

Digital companies use information and knowledge to perform *core* activities, tasks, and services that allow companies to directly create and appropriate value and *complementary* activities, tasks, and services that support the execution of the core ones.⁸⁸ In the Vodafone case, the digital transformation had the objective of redefining the way the company listens to and understands end customers. The information is used to train AI in customer care and provide cognitive services. Along with the transformation of the core activities, the digital transformation also provided Vodafone with the opportunity to use the new and in-depth knowledge about customers to offer personalized products and services. Similarly, ABB exploits information and knowledge to tailor solutions and offer savings to customers, execute predictive maintenance, and provide automatic reordering. The digital transformation project also allowed ABB to offer accurate assistance to customers as a complementary activity. Finally, CNH Industrial uses the information and knowledge to improve activities (such as fleet management, failure prediction, remote vehicle monitoring) and to enhance automation capabilities.

Customers

The last building block of our framework involves the customers for whom the digital company creates value, and it has been strongly stressed in prior studies.⁸⁹ We distinguish between *existing* and *new* customers for the digital transformation project. CNH Industrial strengthened its relationships with the existing customer base as a result of their closer connection in the new business model, aiming to make these relationships mutually more valuable. The availability of new data and information also enabled new customer profiles to be addressed. ABB added new customers to its existing base, such as OEMs and distributors. Customers can be both internal and external to the company. Vodafone's digital transformation project, for example, is targeted at internal customers, such as other Vodafone business units, and external customers, such as end users.

The framework's building blocks resemble the business model elements proposed in the literature.⁹⁰ The framework (Figure 1) organizes and illustrates the building blocks related to a firm's value proposition and market segments as the scope and the customers blocks. The central part of the framework (data, data platform, AI, information and knowledge, people, and partners blocks) contains the value chain structure of the digital business model. The transformed activities, tasks, and services block represents the firm's capability to extract rent from the value created. Accordingly, this block is connected with the mechanisms used by the firm to appropriate the value provided to the customers. Finally, the relationships among the different elements are defined through the processes and procedures block and by the structure itself of the framework. In the appendix, we illustrate how the framework is applied to describe the three cases presented in this study.

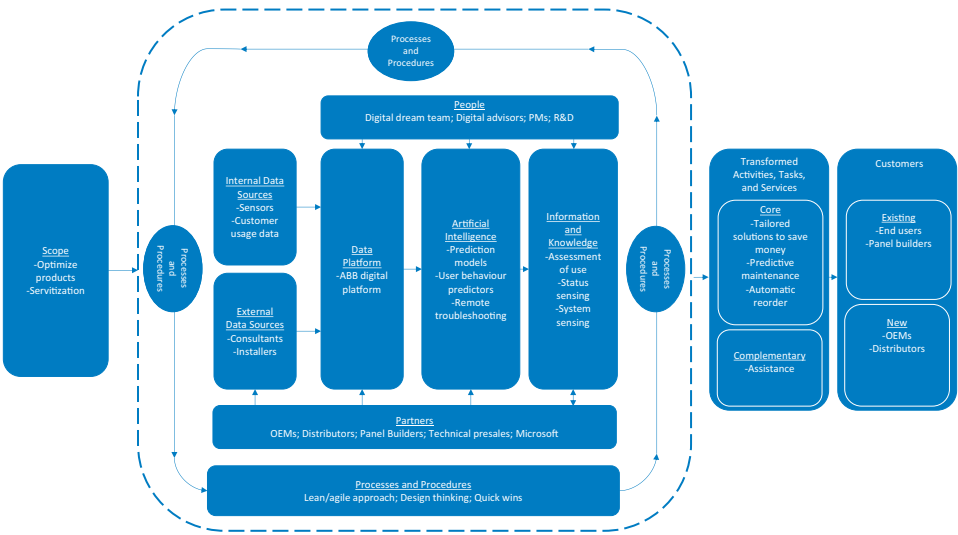
Conclusion

To understand how digital transformation can be effectively implemented, we have identified the building blocks underlying the implementation of a digital transformation strategy. Through the analysis of ABB, CNH Industrial, and Vodafone, three companies that successfully undertook the digital transformation journey assisted by Microsoft, we constructed a framework that can support companies in digitally transforming their businesses and creating a strong and consistent connection between strategy formulation and strategy implementation. More specifically, this framework serves as an actionable guide that can help companies navigate the challenges associated with the implementation of a digital transformation strategy. This process requires that companies rethink and revamp their business models in order to reduce risk and uncertainty. In this regard, our framework can serve as a checklist to ensure that none of the key elements composing the strategy is neglected when senior executives engage in digital strategy implementation.

Appendix

Here, we report on how the framework applied to the three cases analyzed: ABB (Figure A1), CNH Industrial (Figure A2), and Vodafone (Figure A3).

FIGURE A1. The framework applied to the ABB case.



Note: OEMs = original equipment manufacturers; PMs = product managers; R&D = research and development.

FIGURE A2. The framework applied to the CNH Industrial case.

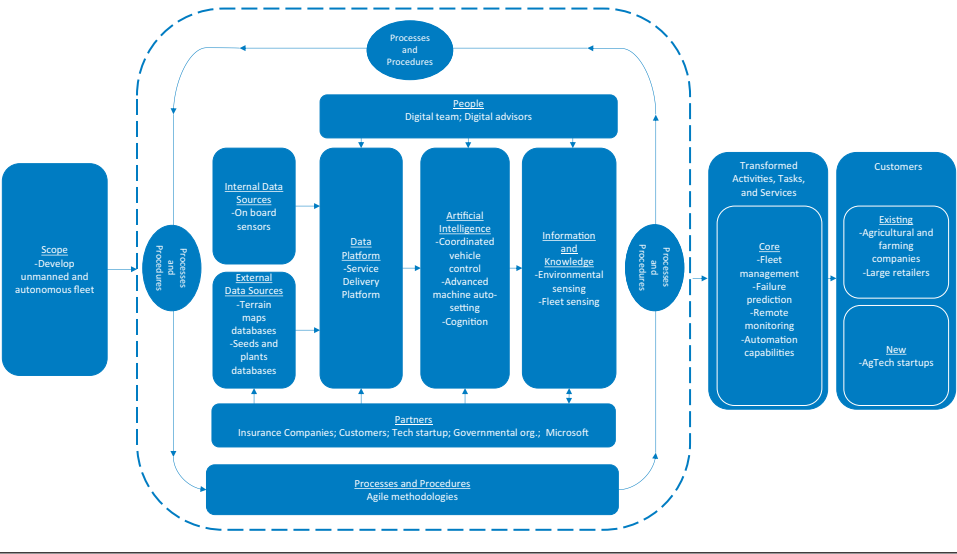
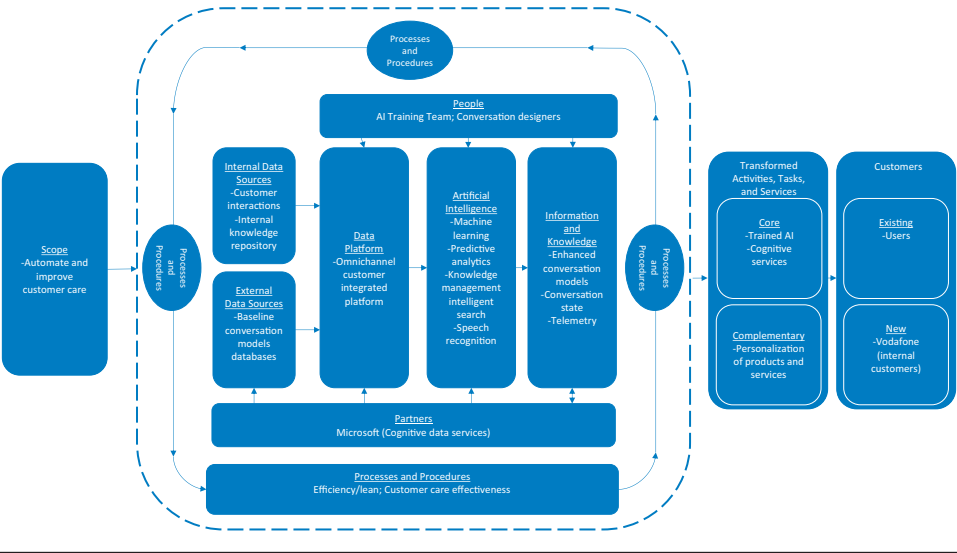


FIGURE A3. The framework applied to the Vodafone case.



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